Precautionary Pricing

Mahdi Ebsim*

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Abstract

Firms set prices and production quantities before demand is realized, exposing them to the risk of excess inventories or stock-outs. I develop a model in which risk-averse firms choose prices and inventories under idiosyncratic demand uncertainty. Precautionary behavior generates dispersion in prices and markups even with constant demand elasticity, as firms trade off insuring against surpluses and insuring against stock-outs. Quantitatively, the model replicates observed relationships between volatility, inventory investment, and markup dispersion in U.S. data. During supply chain disruptions, heightened stock-out risk raises aggregate markups and delays recovery, revealing how firm-level risk aversion amplifies aggregate fluctuations.

*New York University. Email: msebsim@gmail.com

1 Introduction

In most industries, firms must determine production quantities and prices prior to the realization of demand. Such timing exposes them to idiosyncratic uncertainty: unexpected surpluses of inventories when demand is weak or costly stock-outs when demand is strong. These risks are particularly salient for firms introducing new or seasonal products, where pricing and production decisions are taken in advance of sales. Despite the ubiquity of this environment, the role of idiosyncratic risk and risk aversion in firms' price-setting behavior remains largely unexplored in the literature.

This paper develops a model of precautionary pricing in which risk-averse firms choose prices and inventories before demand is realized. Firms face idiosyncratic, persistent shocks to demand and cannot insure against the resulting fluctuations in revenues. As a consequence, they behave as if they were insuring against future shortages or surpluses. This precautionary motive generates dispersion in prices and markups, even though the elasticity of substitution across goods is constant. Firms with weaker balance sheets or greater exposure to demand volatility set higher markups to mitigate the risk of stock-outs, whereas financially unconstrained firms adopt lower prices and higher inventory levels.

The model introduces a trade-off between insuring against the risk of unsold inventories and insuring against the risk of stock-outs. Firms that emphasize insurance against rationing of demand choose higher markups and lower inventories. Aggregating across firms, these heterogeneous responses give rise to variation in markups and utilization rates, leading to misallocation and losses in aggregate productivity. I further show that during a supply chain disruption, which raises the idiosyncratic risk of stock-outs, aggregate markups increase and recovery is delayed relative to an economy with risk-neutral firms. Thus, firm-level risk aversion and incomplete insurance can amplify and prolong supply-side crises.

Quantitatively, the model is calibrated at an annual frequency to match moments on the volatility of sales and inventories from Compustat, the prevalence of stock-outs documented in the literature, and the observed inequality in entrepreneurial wealth. The model reproduces the dispersion of markups and inventory behavior observed in the data. In the empirical analysis, using a panel of U.S. nonfinancial corporations from 2001 to 2019, I show that firms facing greater idiosyncratic volatility in stock returns invest less in inventories, and that this effect is amplified when executives hold equity stakes in their firms. Moreover, variation in firm-specific volatility and managerial exposure accounts for 10 to 25 percent of observed markup dispersion.

1.1 Related Literature

This paper contributes to three strands of literature. First, it contributes to the literature on inventory dynamics and their implications for macroeconomics. Papers such as Bils and Kahn (2000), Khan and Thomas (2007), and Wen (2011) study the role of inventories in the amplification of macro shocks. Relative to this literature, I introduce a mechanism through which idiosyncratic risk and risk aversion jointly shape price-setting and inventory policies. In addition, I follow Alessandria et al. (2023) in studying supply chain crises, modeling the crisis as an increase in the probability of shipment delays.

Distinct from but related to the inventories literature is Flynn et al. (2025). They focus on firms committing to a production schedule given realizations of demand. Although dynamic pricing exists, in the goods market both pricing and output decisions are commonly predetermined, and price rigidity coexists with significant inventory holdings.[1] Hence, I take the starker view that firms are restricted to a single price and quantity rather than a locus of prices and quantities.

Second, it contributes to the literature on firm dynamics and markups. Previous work such as Edmond et al. (2023), Baqaee et al. (2021), and De Loecker et al. (2020) emphasizes the role of markup dispersion among firms. Whereas size-dependent markups are common in this literature, I provide a micro-foundation for markup dispersion arising from incomplete insurance rather than from heterogeneity in demand elasticities, technological differences, or firm size.

Third, it contributes to the literature on firm dynamics and risk. David et al. (2022) provide a framework linking misallocation in capital to risk rather than financial frictions. Similarly, Boar et al. (2022) study a model where misallocation in product markets is due primarily to risk. Relative to these two papers, I introduce pricing and inventory choice as separate policies and study the implications of risk as a primary friction. Finally, Di Tella et al. (2025) study optimal taxation when markups are due to risk. Unlike their paper, I disentangle the sources of risk to which firms are exposed, arising from stock-outs and surpluses, and show that both contribute to markup variation.

2 Model

Time is discrete. The economy is populated by a representative worker, a final good producer, and a unit mass of firms. The representative worker supplies labor and saves or borrows in

[1] The same can be said for services. For instance, restaurants choose prices for their menus in advance. In this case, inventories are not held in the final good but instead in the raw inputs that are used to produce the services.

risk-free bonds. They supply labor to intermediate good firms, which use the labor along with capital to produce inventories for differentiated goods. Sales of inventories are combined by a final good producer. Intermediate good firms are not allowed to borrow and must price and order their inventories before demand for their product is realized. Their firm-specific demand is subject to persistent idiosyncratic shocks. There is no aggregate uncertainty.

2.1 Representative Worker

The representative worker's objective is to maximize their lifetime utility. They discount time with $\beta_w \in (0,1)$ and have period utility $U(C_w, L_w)$ over consumption C_w and labor L_w. Their problem, stated recursively, is

$$V_{w,t}(B_w) = \max_{L_w, B'_w} U(C_w, L_w) + \beta_\omega V_{w,t+1}(B'_w)$$
$$C_w + B'_w \leq (1 + r_t)B_w + W_t L_w$$

They face interest rate r_t and wages W_t. They can borrow subject to the natural borrowing limit. I assume period utility is CRRA with separable disutility from labor, that is, $U(C, L) = \frac{C^{1-\gamma}}{1-\gamma} - \Psi\frac{L^{1+\eta}}{1+\eta}$, with coefficient of relative risk aversion γ and inverse Frisch elasticity η. Therefore, labor and savings choices are such that the following first-order conditions hold

$$W_t = \Psi C_{w,t}^{\gamma} L_{w,t}^{\eta}$$
$$\beta_w \left(\frac{C_{w,t}}{C_{\omega,t+1}}\right)^{\gamma} = \frac{1}{1 + r_{t+1}}$$

2.2 Final Good Firm

The final good firm combines output y_i for firms $i \in [0,1]$ to produce Y. They have CES technology such that

$$Y = \left(\int \theta_i^{\frac{1}{\sigma}} y_i^{\frac{\sigma-1}{\sigma}} di\right)^{\frac{\sigma}{\sigma-1}}$$

where θ_i is a demand shock for firm i. Intermediate good firms sell their output at prices p_i and supply goods up to their available inventories, I_i. As the final good firm is perfectly competitive, their problem is

$$\min_{y_i} \left\{ \int p_i y_i di | Y = \left(\int \theta_i^{\frac{1}{\sigma}} y_i^{\frac{\sigma-1}{\sigma}} di\right)^{\frac{\sigma}{\sigma-1}}, y_i \leq I_i \quad \forall i \right\}$$

Demand for good i is:

$$y_i = Y\left(\frac{p_i + \lambda_i}{P}\right)^{-\sigma}\theta_i$$

where λ_i is the multiplier if firm i experiences a stock-out, which can be expressed as

$$\lambda_i = \max\left\{P\left(\frac{I_i}{Y}\right)^{-\frac{1}{\sigma}}\theta^{\frac{1}{\sigma}} - p_i, 0\right\}$$

The aggregate price index P is

$$P^{1-\sigma} = \int \theta_i \left(p_i + \lambda_i\right)^{1-\sigma} di$$

The aggregate price index P is not equivalent to the CPI, as it is in standard CES models. Let $P_{cpi} = \int p_i \frac{y_i}{Y} di$. We can relate the CPI and P using

$$P = P_{cpi} + \int \lambda_i \frac{I_i}{Y} di$$

Differences between the CPI and the price index are due to the covariance between λ_i and I_i. The aggregate price index is larger if firms with low levels of inventories are rationing their demand. Going forward, I set the aggregate price index as numéraire, $P = 1$.

2.3 Intermediate Good Firms

Intermediate good firms face restrictions in selling their goods. They must set prices and choose output one period in advance. As a consequence, their choices are subject to demand uncertainty. A firm enters the period with prices p_t, inventories of their good I_t, and savings a_t, all determined at $t-1$. At the beginning of the period, demand θ_t is realized, and demand for the firm's good is $y_t = Y_t p_t^{-\sigma}\theta_t$. They can earn at most $p_t \min\{y_t, I_t\}$ in revenue. They can end up with a surplus in inventories or ration demand depending on realizations of θ_t.

At the end of period t, they produce new inventories I_{t+1} by hiring labor ℓ_t at wage W_t and producing using CRS technology. They save in bonds a_{t+1} subject to a no-borrowing constraint. Lastly, they set tomorrow's price p_{t+1}. A fraction $(1 - \delta_I)$ of unsold inventories

can be liquidated. The firm's problem, stated recursively, is

$$\begin{aligned} V_t(\theta, m) = \max_{p', I', a'} \; & u(c) + \beta \mathbb{E}_t \left[V_{t+1}(\theta', m'(p', I', a')) \right] \\ & c \leq m - a' - W_t I' \\ & m' = p' \min\{y', I'\} + (1 + r_{t+1}) a' + (1 - \delta_I) \min\{I' - \min\{y', I'\}\} \\ & y' = Y_{t+1} (p')^{-\sigma} \theta' \\ & 0 \leq a' \end{aligned}$$

Their period utility is $u(c_t) = \frac{c_t^{1-\gamma}}{1-\gamma}$. Firm demand evolves according to

$$\log \theta' = \rho_\theta \log \theta + \nu'$$

where

$$\nu' \sim \begin{cases} N(0, \sigma_\theta) & \text{with prob. } \chi \\ N(0, \tilde{\sigma}_\theta) & \text{with prob. } (1 - \chi) \end{cases}$$

Demand evolves according to a log-autoregressive process with innovations that are a mixture of two normals, where $\tilde{\sigma}_\theta > \sigma_\theta$, and are mixed with probability $\chi \in (0, 1)$. This will generate significant risk in pricing since demand will be subject to fat-tailed shocks. The firm's state is their demand and cash on hand m, which consists of sales of the intermediate good, savings, and liquidated inventories. Let $F(\theta, m)$ be the distribution of intermediate good firms in the space of (θ, m).

2.4 Discussion

Timing. Firms are restricted to choosing output and prices one period in advance. Time-to-build is natural for capital and inventories. It is well established that, at business cycle frequency, prices are sticky. In the environment I present, firms introduce new prices through new products. One should view these as single-product firms, which introduce a new price and product. This is natural in consumer goods such as electronics and clothing. Certain goods are "in-season" for a period of time, and then new iterations are introduced with new prices. Argente et al. (2024) show that prices fall for products over their lifetime and new products are introduced with newer, higher prices.

Liquidation. I allow the firm to claim a fraction of unsold inventories for consumption. Compared to other models (e.g., Kryvtsov and Midrigan (2013)) I am allowing reversibility in inventories. The crucial difference in this setting is the frequency. I consider a model with

lower frequency. I calibrate the model to an annual frequency. At business cycle frequency, it is natural to think that inventories are irreversible. However, at annual frequency this assumption is unlikely to be accurate. Markets for liquidation and fire sale of unsold inventories are well established.[2] I abstract from characterizing this market and assume firms take a significant haircut on their unsold inventories.

Risk Aversion. I assume a firm is managed by a single entrepreneur who owns the firm outright. In reality, firms are managed by multiple managers. In the case of public firms, these managers share a stake in the firm's earnings alongside other shareholders. I will show in the empirical analysis that public firms exhibit evidence of risk aversion through their inventory investment. When firms' executives have a higher ownership stake, their inventory investment is attenuated relative to firms with unexposed executives. This result replicates a well-known result for capital investment found in Panousi and Papanikolaou (2012). A model with risk-neutral firms will imply prices and inventories are only a function of exogenous variation in demand. The data suggest that variation in inventories is strongly related to the firm's balance sheet and the exposure of firms' executives to earnings. Both these facts are consistent with risk aversion but not consistent with risk neutrality.

2.5 Analysis

Policies. The firm's first-order condition for savings a' is

$$u'(c) = \beta(1 + r_{t+1} + \mu)\mathbb{E}_t\left(V_{m,t+1}\right)$$

where the multiplier on $a' \geq 0$ is μ and $V_{m,t+1} \equiv \frac{\partial V_{t+1}(\theta,m)}{\partial m}$. The first-order condition for new inventories I' is

$$u'(c)W_t = \beta\mathbb{E}_t\left[V_{m,t+1}\left\{\mathcal{X}_{t+1}p' + (1-\mathcal{X}_{t+1})(1-\delta_I)W_{t+1}\right\}\right]$$

where $\mathcal{X}_t = \mathbb{I}\left(y' \geq I'\right)$, an indicator of a stock-out. To simplify expressions going forward, define $\hat{\mathbb{E}}_t(X) \equiv \frac{\mathbb{E}[V_{m,t+1}X]}{\mathbb{E}[V_{m,t+1}]}$, which is the expectation operator taken under the risk-neutral measure. Combining the two optimality conditions,

$$\left(1 + r_{t+1} + \mu\right)W_t = \hat{\mathbb{E}}_t\left[\mathcal{X}_{t+1}p' + (1-\mathcal{X}_{t+1})(1-\delta_I)W_{t+1}\right]$$

[2]NAICS sector 561990 includes inventory liquidators.

On the left-hand side $(1 + r_{t+1} + \mu)W_t$ is the marginal cost of inventories. It includes the labor hired, W_t, the opportunity cost of saving instead, r_{t+1}, and the cost of financial frictions μ. On the right-hand side is the marginal benefit of inventories. In the case of stock-outs, i.e. $\mathcal{X}_{t+1} = 1$, one earns p' in revenues. Also, if there is no stock-out, then one earns the liquidation value of the inventories.

The first-order condition for p' is

$$p' = \frac{\sigma \hat{\mathbb{E}}_t \left[y'(1 - \mathcal{X}_{t+1}) \right]}{\hat{\mathbb{E}}_t \left[(\sigma - 1) y'(1 - \mathcal{X}_{t+1}) - I' \mathcal{X}_{t+1} \right]} W_{t+1}(1 - \delta_I)$$

While this is close to an expression of prices as a markup over a marginal cost of production, $W_{t+1}(1-\delta_I)$ is not the marginal cost but rather the liquidation value. We can arrive at a familiar pricing expression by combining this optimality condition with the one for inventories, such that

$$p' = \underbrace{\frac{\hat{\sigma}}{\hat{\sigma} - 1}}_{\text{markup}} \underbrace{W_t(1 + r_{t+1} + \mu)}_{\text{marginal cost}} \tag{1}$$

where markups are a Lerner index of the "effective" demand elasticity $\hat{\sigma}$. This elasticity can be expressed as

$$\hat{\sigma} = \sigma \frac{\hat{\mathbb{E}}_t \left(y' | y' \leq I' \right)}{\hat{\mathbb{E}}_t \left(\min \left\{ y', I' \right\} \right)} \tag{2}$$

which is the elasticity of substitution σ under CES scaled by the expected output conditional on not stocking out relative to expected sales, all under the risk-neutral measure. Note that $\hat{\sigma} \leq \sigma$; hence, markups are higher due to demand uncertainty.

Importantly, in this environment markups vary with respect to wealth. Why? Note that $\hat{\sigma}$ varies with respect to (θ, m), the idiosyncratic state of the firm. This is because expectations are conditioned on $V_m(\theta, m)$. By the envelope theorem: $V_m(\theta, m) = u'(c)$, hence expectations are weighted by marginal utility of consumption. The covariance between consumption and outcomes will be relevant for inventory demand and pricing. If output is lower in bad states of the world, i.e. $\hat{\mathbb{E}}_t \left(y' | y' \leq I' \right) \leq \mathbb{E}_t \left(y' | y' \leq I' \right)$, then $\hat{\sigma}$ falls, and markups increase. The same holds if the probability of a stock-out is correlated with low consumption.

Markups in this environment reflect (1) market power due to monopolistic competition, (2) variable utilization as demand is uncertain, and (3) risk. The risk occurs because shortfalls in demand are correlated with low consumption. Due to this, risk-averse firms will choose to price with a markup to be compensated for risk.

Risk Neutrality. To understand the relevance of risk, it is useful to compare it with the risk-neutral benchmark. Suppose p' and I' are chosen by a firm with risk-neutral preferences or in an environment with complete markets. A firm that chooses price p' and inventories I' experiences a stock-out if $y' \geq I'$. Substituting in demand y', stock-outs occur if:

$$Y_{t+1} (p')^{-\sigma} \theta^{\rho} \exp(\nu') \geq I'$$

Therefore, stock-outs occur if $\nu \geq \bar{\nu}$ where

$$\bar{\nu} \equiv \log \left(\frac{I'}{Y_{t+1} (p')^{-\sigma} \theta^{\rho}} \right)$$

This is the cutoff for the innovation in demand such that any innovations above this level lead to stock-outs. With risk neutrality, the first-order conditions of the firm no longer involve the risk-neutral measure. Combining the conditions for inventories and prices, the firm's policy then is a cutoff $\bar{\nu}$. This is a similar result to papers on inventory dynamics and the business cycle such as Kim (2021) and Wen (2011). The choice of $\bar{\nu}$ is isomorphic to the probability of stock-outs. Implicitly, the risk neutral firm chooses the probability of stock-outs and then backs out the unique pair of p' and I' to generate such outcomes. In addition, note that $\bar{\nu}$ is state-independent. In fact, it is only a function of wages, elasticity of demand σ, and distribution of the innovation ν. Therefore, the ex-ante stock-out probability without risk aversion is state-independent. As ex-ante stock-out probabilities are constant across firms, we have the following implication:

Proposition 1. *If a firm is risk neutral, then prices, p', are constant with respect to (θ, m).*

As a corollary, markups are constant and do not vary across risk-neutral firms. This is because costs are identical. In their environment, the only difference that demand θ generates is a difference in sales. As a consequence, in the baseline model, with risk-averse firms, any and all markup or price dispersion is due to differences in risk exposure. Variation in markups will be purely due to risk. Variation in prices will also include a portion due to costs of borrowing passed on to the consumer.

Aggregate Markup. Define the aggregate markup $\mathcal{M}_t$ implicitly as

$$\frac{W_t(1 + r_t)L_{w,t}}{Y_t} = \frac{1}{\mathcal{M}_t}$$

which admits a harmonic average representation:

$$\frac{1}{\mathcal{M}_t} = \int \frac{p_{i,t} \min\{y_{i,t}, I_{i,t}\}}{Y_t} \frac{1}{m_{i,t}} \frac{1}{1+\mu_{i,t}/(1+r_t)} \frac{1}{\omega_{i,t}} di$$

where $m_{i,t} = \frac{\hat{\sigma}_{i,t}}{\hat{\sigma}_{i,t}-1}$ is the markup and $\omega_{i,t} = \frac{\min\{y_{i,t}, I_{i,t}\}}{I_{i,t}}$ is the utilization rate of inventories. Markup dispersion will lead to misallocation loss, similar to other models with markup dispersion. In addition, dispersion in utilization rate contributes to misallocation. Finally, financial frictions contribute to misallocation. The borrowing wedge $\mu_{i,t}$ is present in the markup expression. Hence, variation in financing costs will impact aggregate markups.

Aggregate Productivity. Define aggregate productivity Z_t implicitly as $Y_t = Z_t I_t$ where I_t is the aggregate stock of inventories. Aggregate productivity can be expressed as

$$\frac{1}{Z_{t+1}} = \frac{\sigma-1}{\sigma} \int \frac{\hat{\mathbb{E}}_t \left(\min\{y_{i,t+1}, I_{i,t+1}\}\right)}{Y_{t+1}} \frac{1}{\hat{\mathbb{P}}\left(y_{i,t+1} \geq I_{i,t+1}\right)} \frac{1 - \frac{\sigma}{\sigma-1}\frac{1}{m_{i,t+1}}\frac{1-\delta_I}{1+r_{t+1}+\mu_{i,t+1}}}{1 - \frac{1}{m_{i,t+1}}\frac{1-\delta_I}{1+r_{t+1}+\mu_{i,t+1}}} di$$

Aggregate productivity is also a harmonic average. Within the integral the first term is the expected sales of the firm under the risk-neutral measure. The second term is the inverse of the stock-out rate, also under the risk-neutral measure. The final term in the integral reflects the differences between markups from the frictionless level under CES, $\frac{\sigma}{\sigma-1}$. To understand the terms above more concretely, I define $s_{i,t} = \frac{\min\{y_{i,t}, I_{i,t}\}}{Y_t}$ as the output share. In addition, define

$$\zeta_{i,t} = \frac{1 - \frac{\sigma}{\sigma-1}\frac{1}{m_{i,t+1}}\frac{1-\delta_I}{1+r_{t+1}+\mu_{i,t+1}}}{1 - \frac{1}{m_{i,t+1}}\frac{1-\delta_I}{1+r_{t+1}+\mu_{i,t+1}}}$$

$$\xi_{i,t} = \frac{Cov_{i,t}\left(u'(c_{i,t+1}) \min\{y_{i,t+1}, I_{i,t+1}\}\right)}{E_{i,t}\left[u'(c_{i,t+1})\right] E_{i,t}\left[\min\{y_{i,t+1}, I_{i,t+1}\}\right]}$$

Here $\zeta_{i,t}$ is a wedge reflecting the gap between realized markups $m_{i,t}$ and the frictionless counterpart. The wedge $\xi_{i,t}$ is the risk firms are exposed to, the covariance between their marginal utility of consumption and realized sales. If a firm has full insurance, we expect $\xi_{i,t} = 0$, and otherwise $\xi_{i,t} < 0$. We can express aggregate productivity as

$$\frac{1}{Z_{t+1}} = \frac{\sigma-1}{\sigma} \int s_{i,t} \left(1+\xi_{i,t}\right) \frac{\zeta_{i,t+1}}{\hat{\mathbb{P}}\left(y_{i,t+1} \geq I_{i,t+1}\right)} di$$

That is, aggregate productivity is an output-weighted harmonic average of important cross-sectional wedges. To understand the sources of misallocation, I begin with the tractable

case where inventories fully depreciate, $\delta_I = 1$. This means that $\zeta_{i,t} = 1$ for all i. If firms can fully insure against uncertain demand, then, using Proposition 1, we know that the ex-ante probability of a stock-out is common across firms. Let Z_t^* be productivity in this counterfactual and let $\mathbb{P}^*$ be the stock-out probability common to all firms in that equilibrium. If we hold the distribution of $s_{i,t}$ constant, it follows that

$$\log\left(\frac{Z_{t+1}^*}{Z_{t+1}}\right) \approx \mathbb{E}\left(\log\frac{\mathbb{P}^*}{\hat{\mathbb{P}}\left(y_{i,t+1} \geq I_{i,t+1}\right)}\right) + \mathbb{E}\left(\log(1+\xi_{i,t})\right) \tag{3}$$

$$+ \frac{1}{2}\left[\mathbb{V}ar\left(\log\frac{\mathbb{P}^*}{\hat{\mathbb{P}}\left(y_{i,t+1} \geq I_{i,t+1}\right)}\right) + \mathbb{V}ar\left(\log(1+\xi_{i,t})\right)\right] \tag{4}$$

$$- \mathbb{C}ov\left(\log\frac{\mathbb{P}^*}{\hat{\mathbb{P}}\left(y_{i,t+1} \geq I_{i,t+1}\right)}, \log(1+\xi_{i,t})\right) \tag{5}$$

A consequence of this result is that risk misallocates through dispersion in ex-ante stock-out rates. Equation (3) is a second order expansion of the productivity gap between full insurance and the benchmark model. The first term is the average distance between the insured firms' stock-out rate and the stock-out rate in the incomplete-markets economy. The second term is the risk wedge; the average risk wedge increases losses as it implies more output is allocated to exposed firms, which we expect to price with a significant markup. The third and fourth terms are the dispersion in stock-out rates and risk wedge. More dispersion in stock-out rates or risk wedge leads to greater misallocation loss due to incomplete insurance.

This provides a prediction for the degree of misallocation due to risk as a function of two margins: an extensive margin reflecting the stock-out rate and an intensive margin reflecting how much of the inventories is sold. For the extensive margin, it is summarized by the mean and variance of the gap between stock-out rates under complete insurance and stock-out rates faced by risk-averse firms under their risk-neutral measure. Due to Proposition 1, I can conveniently solve for $\mathbb{P}^*$ without prices as it is a function only of model primitives. The intensive margin is similarly summarized by the first two moments.

Very importantly, the two margins will push misallocation in opposite directions. The extensive margin will increase misallocation. As I will show in the following sections, this margin is associated with significant risk markups. The intensive margin on the other hand will be negative, hence reducing misallocation. Why? The logic is similar to Ebsim (2025): risk markups are allocated to firms with high marginal utility, hence allowing them to mitigate their risk through profits.

2.6 Equilibrium

My analysis will focus on the stationary equilibrium. To that end:

Definition 1. *A stationary equilibrium is an interest rate r, output Y, and wage W such that given r, Y, and W: (1) B_w, L_w are optimal for the worker. (2) $p'(\theta, m)$, $I'(\theta, m)$, and $a'(\theta, m)$ solve the firm's Bellman equation, (3) Goods market clears*

$$Y = \int c(\theta, m) dF(\theta, m) + C_w$$

(4) Asset market clears

$$B_w + \int a'(\theta, m) dF(\theta, m) = 0$$

(5) Labor market clears

$$L_w = \int I'(\theta, m) dF(\theta, m)$$

(6) Distribution $F(\theta, m)$ is consistent with firm policies

I assume that $\beta < \beta_w$; therefore, in equilibrium $r = \frac{1}{\beta_w} - 1$. By Walras' law, Y and W which clear goods and labor markets will automatically clear the asset market.

3 Quantitative Analysis

In this section, I present the quantitative results of the model.

3.1 Calibration

The model is calibrated at an annual frequency using cross-sectional information on stock-out rates from the literature and moments on sales and inventories from Compustat. First, some parameters are fixed and presented in Table 1. I set the elasticity of demand to $\sigma = 3$, following Hsieh and Klenow (2009). I set the interest rate to $r = 0.02$ to reflect low interest rates in the United States since 2000. This implies that $\beta_w = 0.98$. I set the coefficient of relative risk aversion $\gamma = 2$ and the inverse Frisch elasticity of labor $\eta = 2$. The disutility of labor Ψ is set such that aggregate labor is 0.6. Finally, the persistence of demand is $\rho = 0.794$ using the annualized persistence of TFPR estimate from Foster et al. (2008).

The remaining parameters, chosen to match moments drawn from the data, are the discount rate of firms, β, the depreciation rate of inventories δ_I, and parameters governing

Table 1: Fixed Calibrated Parameters

Parameter	Value	Description	Source
σ	3	Demand Elasticity	Hsieh and Klenow (2009)
β_w	0.98	Worker's Discount Factor	2% annual interest rate
γ	2	Risk Aversion	–
ρ	0.794	Persistence of $\log\theta$	Foster et al. (2008)
η	0.5	Inverse Frisch Elasticity	Arellano et al. (2019)
Ψ	1.78	Labor Disutility	$L_w = 0.6$

Table 2: Fitted Calibrated Parameters

Parameter	Value	Description
β	0.87	Firm's Discount Factor
δ_I	0.95	Inventory Depreciation Rate
σ_θ	0.09	Std. of $\log\theta$
$\tilde{\sigma}_\theta$	0.75	Std. of $\log\theta$
$1-\chi$	0.25	Prob. of bad state

Table 3: Targeted Moments

Moment	Model	Data
Gini	0.72	0.78
Stock-out Rate	0.69	0.72
$SD\,(\%\Delta p \cdot y)$	0.31	0.29
$SD\,(\%\Delta I)$	0.34	0.40
$Corr\,(\%\Delta I, m)$	0.64	0.68

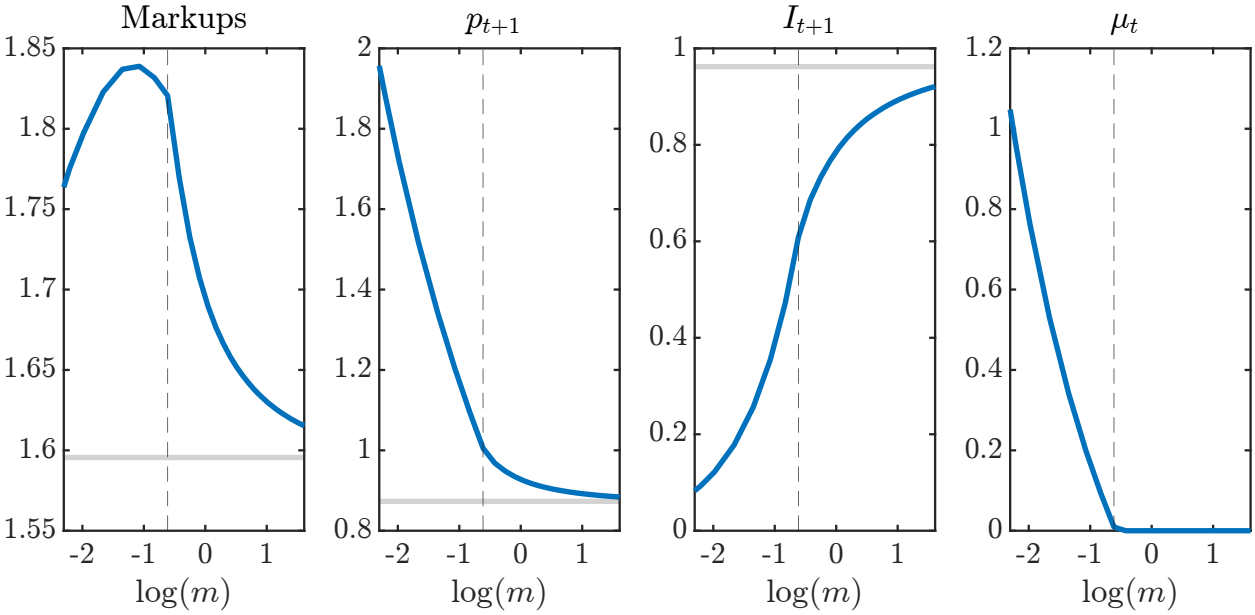
demand shocks (σ_θ, $\tilde{\sigma}_\theta$, and χ). I choose these parameters to match moments capturing the volatility of sales and inventories, the propensity of firms to stock out, and the inequality in wealth among firms. To capture inequality, I target a Gini coefficient of wealth of 0.78 for entrepreneurs in the U.S. from Boar and Midrigan (2023). To capture volatility of sales and inventories, from Compustat, I target the standard deviation of annual sales and inventory

growth. Also from Compustat, to capture how risk affects inventory accumulation through wealth, I target the correlation between inventories and cash.

Lastly, I need a target moment for stock-out rates. Cavallo and Kryvtsov (2023) document a weekly stock-out rate during the COVID-19 pandemic for U.S. retailers in a number of product categories. The weekly stock-out rates range from 2 percent to 10 percent for the pre-crisis trend, which I treat as the long-term average. Since I model slower moving inventories, extending this rate to an annual frequency risks biasing results. I assume that stock-out rates are constant within a quarter and that firms are flexible after a quarter. They either liquidate inventories or draw them down completely. The target stock-out rate then is the weekly rate reported at a quarterly frequency, that is $0.02^{1/12} = 0.72$ or 72 percent of firms experience a stock-out. Table 2 presents values for these parameters. Table 3 presents the model's moments from simulation and compares them to the targets.

3.2 Firm Policies

Figure 1: Firm Policies for Median θ



Prices and Inventories. Figure 1 plots firm policies for markups, prices, and inventories and the multiplier on the borrowing constraint in the last panel. This is plotted for the median θ and against cash on hand m. The vertical dashed line identifies the level in m where $\mu_t > 0$. To the left of the line, $a' = 0$, and to the right, $a' > 0$. The horizontal gray lines are the policies of the risk-neutral firm facing the same equilibrium prices.

First, inventories are significantly low for firms in a financially poor state. For firms with a binding constraint, the gap to the complete insurance policy is significant. However, as firms face uninsurable risk, inventory policy is far from the complete insurance level for levels of cash on hand where they are not presently financially constrained. This occurs because investment in inventories is risky. Not only is output volatile, but, because of the timing, surpluses in inventories also lead to significant downside risk.

Second, note that prices are higher for low cash-on-hand firms. In addition, prices are higher than if set with complete insurance. The variation in prices is greater when $\mu_t > 0$. As cash on hand increases, prices fall towards the complete insurance counterfactual. However, there is still a significant gap. This occurs because the risk of stock-outs or of the borrowing constraint binding in the future is very costly.

Risk Markups. Markups are significantly higher for risk-averse firms than under the complete insurance benchmark. Markups increase with respect to cash on hand but then fall when a firm is no longer bound by the no borrowing constraint. Markups in this environment are a consequence of risk. Firms with greater levels of demand, θ, will face relatively more downside risk, all else equal. Therefore, we can understand the nature of precautionary behavior in this model by comparing policies for different θ and the associated covariances between marginal utility, stock-outs, and utilization.

Figure 2: Firm Policies Relative to Full Insurance Benchmark by θ

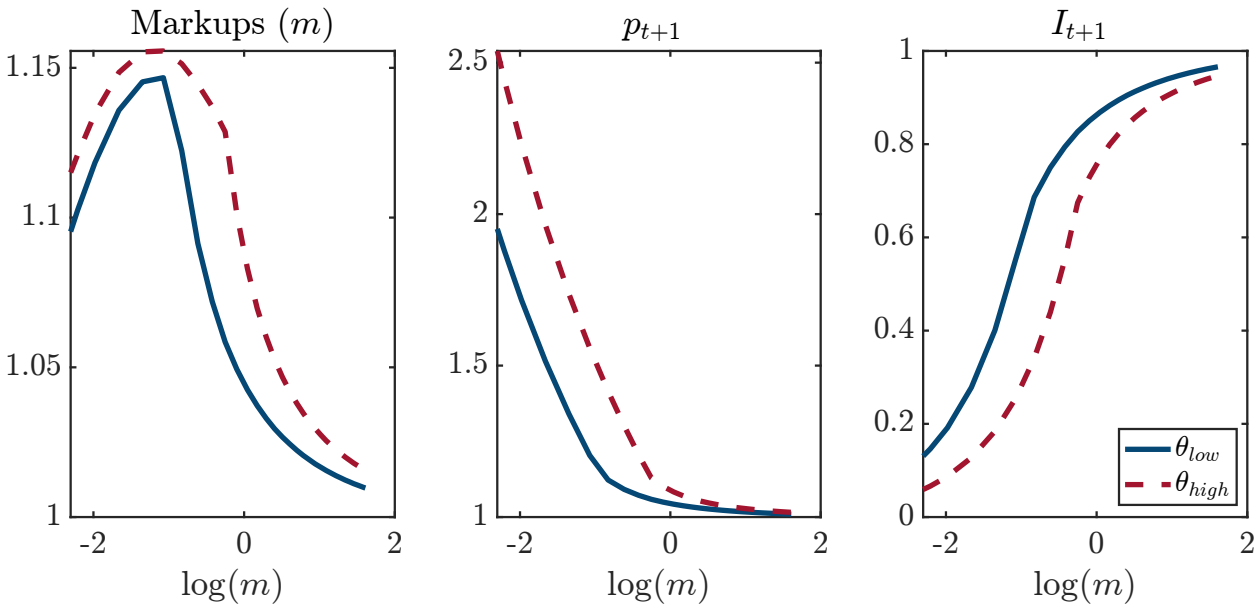


Figure 2 plots the policies for two different levels of θ relative to the complete insurance benchmark. Note that prices and markups increase, and inventories decrease, as θ

increases. Since sales are greater with greater θ, the model still preserves what is sometimes termed "Marshall's Second Law." Nonetheless, markups increase with θ not because demand becomes more inelastic but because of precautionary behavior.

To understand the role of precautionary behavior in generating markups, recall that markups are $\frac{\widehat{\sigma}}{\widehat{\sigma}-1}$ where

$$\begin{aligned}\hat{\sigma} &= \sigma \frac{\widehat{\mathbb{E}}\left(y' | y' \leq I'\right)}{\widehat{\mathbb{E}}\left(\min\left\{y', I'\right\}\right)} \\ &= \sigma \frac{\widehat{\mathbb{E}}\left(\frac{y'}{I'} | y' \leq I'\right)}{\widehat{P}\left(y' \geq I'\right)\left\{1 - \widehat{\mathbb{E}}\left(\frac{y'}{I'} | y' \leq I'\right)\right\} + 1}\end{aligned}$$

Very importantly, the elasticity of substitution varies depending on two margins. First, the extensive margin $\widehat{P}\left(y' \geq I'\right)$, the ex-ante probability of stock-outs under the risk-neutral measure. Second, the intensive margin $\widehat{\mathbb{E}}\left(\frac{y'}{I'} | y' \leq I'\right)$, the expected utilization rate of inventories, also under the risk-neutral measure.

Expressing these two margins without the risk-neutral measure:

$$\begin{aligned}\widehat{P}\left(y' \geq I'\right) &= P\left(y' \geq I'\right) + Cov\left(\frac{u'(c')}{\mathbb{E}(u'(c'))}, \mathbb{I}\left(y' \geq I'\right)\right) \\ \widehat{\mathbb{E}}\left(\frac{y'}{I'} | y' \leq I'\right) &= \mathbb{E}\left(\frac{y'}{I'} | y' \leq I'\right) + Cov\left(\frac{u'(c')}{\mathbb{E}(u'(c'))}, \frac{y'}{I'}\right)\end{aligned}$$

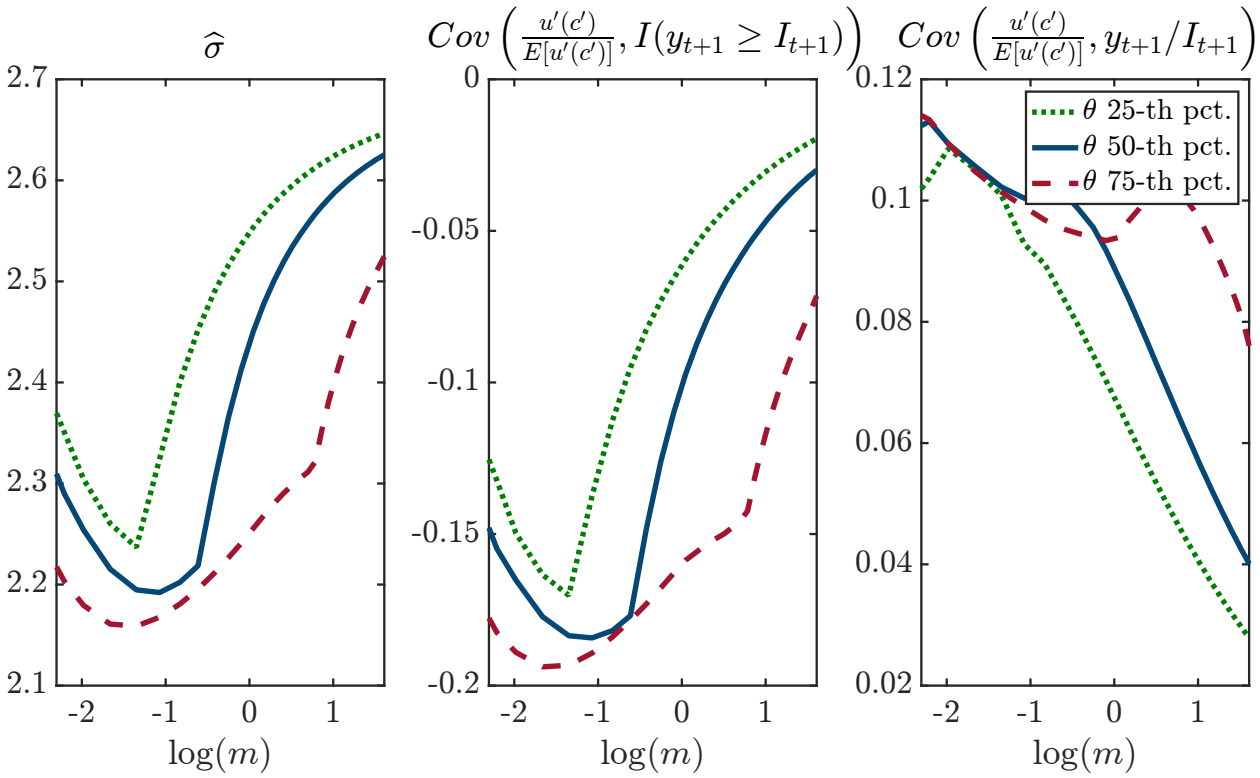
The extensive margin is associated with risk when stock-outs are more likely to occur in states with high marginal utility of consumption. Intuitively, in states where consumption is valuable the opportunity cost of missed profits due to setting prices in retrospect too low or producing too few inventories is costly. In such cases, $Cov\left(\frac{u'(c')}{\mathbb{E}(u'(c'))}, \mathbb{I}\left(y' \geq I'\right)\right) > 0$. This increases $\widehat{P}\left(y' \geq I'\right)$, lowering the elasticity of demand and hence increasing markups. The risk in this margin comes from missing out on valuable sales and rationing demand.

The intensive margin reflects risk if the utilization rate of inventories is low. This margin reflects risk from carrying surplus inventories. Inventories are risky as bad shocks in demand will lead to unsold inventories and liquidated at significant discounts relative to costs. This risk is reflected by $Cov\left(\frac{u'(c')}{\mathbb{E}(u'(c'))}, \frac{y'}{I'}\right) < 0$. Thus, if low utilization coincides with bad states for the firm, the expected utilization rate under the risk-neutral measure is lower. This drives markups higher, all else equal.

These margins are irrelevant under complete insurance; however, the risk-averse manager internalizes them. Therefore, evidence of risk aversion will be avoidance of stock-outs when cash on hand, m, is low, and/or an increase in utilization. Figure 3 plots the elasticity of

demand and the two covariances of interest.

Figure 3: Elasticity of Demand and Risk by θ



In attempting to deal with the covariance of utilization and stock-outs with marginal utility, the firm faces a trade-off. If they reduce expected stock-outs in bad states, they invariably reduce expected utilization and vice-versa. Therefore, results in Figure 3 suggest that firms increase their markups in low risk states by increasing their utilization. As risk increases, presented in Figure 3 as θ at the 75th percentile, the covariance of marginal utility with stock-outs falls and so does the covariance with utilization. Hence, the firm moves in the direction of policy that takes relatively greater precaution against stock-outs versus slack capacity.

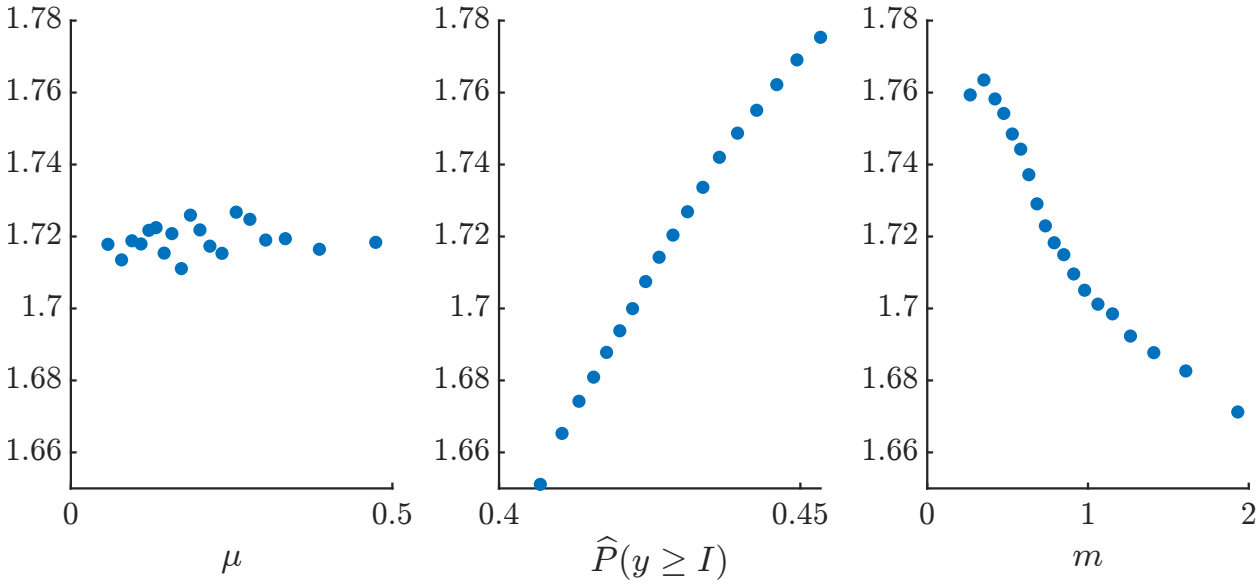
In summary, precautionary behavior in this environment generates markups. These markups are informative about two sources of risk. First, along the extensive margin, the risk of rationing or stocking out of inventories, and hence of missing out on profits. Second, along the intensive margin, the risk of unsold inventories. As the firm's level of risk increases, precautionary behavior moves from self-insurance against intensive margin risk to extensive margin.

Financial Frictions. The borrowing constraint, $a' \geq 0$, plays a role in firm policies. Note that monotonicity of markups changes around the vertical dashed line in Figure 1. The

marginal cost of inventories for a firm is $W(1 + r + \mu)$. The multiplier on the no borrowing constraint, μ, varies across firms and in particular across cash on hand, m. From the definition of markups, holding costs constant, all deviations from the complete insurance benchmark are due to the covariance of stock-outs and output with marginal utility. Hence, contemporaneous financial frictions do not affect markups. The non-monotonicity of markups around when μ becomes positive reflects a change in the main distortion faced by firms. As the financial wealth of firms increases, the cost of financial frictions falls and the prominence of risk as the main distortion rises. Idiosyncratic risk faced by firms is highest when $\mu = 0$. Firms have CRS production but face CES demand. In this environment, markups have similar implications to the risk wedge in Boar et al. (2022) and Di Tella et al. (2025), where firms are perfectly competitive but have DRS production.

Cross-section. To confirm that contemporaneous financial frictions are not crucial for markup dispersion, in Figure 4 I plot binscatters of markups as a function of μ, the multiplier on the no-borrowing constraint, $\widehat{P}(y' \geq I')$ the stock-out rate, under the risk-neutral measure, and cash on hand. In the ergodic distribution, there is very little variation in markups across μ. Instead, significant variation occurs across firms with different stock-out rates and financial wealth.

Figure 4: Markups Binscatter



3.3 Consequences for Productivity

In this section, I present results of the misallocation decomposition in equation (3):

$$\underbrace{\log\left(\frac{Z^*_{t+1}}{Z_{t+1}}\right)}_{2.80\%} \approx \underbrace{\mathbb{E}\left(\log\frac{\mathbb{P}^*}{\hat{\mathbb{P}}\left(y_{i,t+1}\geq I_{i,t+1}\right)}\right)}_{11.07\%} + \underbrace{\mathbb{E}\left(\log(1+\xi_{i,t})\right)}_{-8.49\%}$$
$$+ \underbrace{\frac{1}{2}\left[\mathbb{V}ar\left(\log\frac{\mathbb{P}^*}{\hat{\mathbb{P}}\left(y_{i,t+1}\geq I_{i,t+1}\right)}\right) + \mathbb{V}ar\left(\log(1+\xi_{i,t})\right)\right]}_{0.11\%}$$
$$- \underbrace{\mathbb{C}ov\left(\log\frac{\mathbb{P}^*}{\hat{\mathbb{P}}\left(y_{i,t+1}\geq I_{i,t+1}\right)}, \log(1+\xi_{i,t})\right)}_{-0.11\%}$$

As predicted above, the two margins have opposing implications for misallocation. Losses as a consequence of risk are 2.8 percent. The gap in stock-out rates contributes an 11 percent loss. This is a significant amount and is counteracted partially by the intensive margin, $\xi_{i,t}$, which reduces misallocation by 8.5 percent. The second order effects are small. A significant implication of this outcome is that misallocation is generated primarily by suboptimal stock-out rates of risk-averse firms. Since running out of inventories is costly, especially as exposure to idiosyncratic risk increases, the probability of stock-outs falls. The lower stock-out rates are generated by higher prices, lower inventory demand, and greater markups.

3.4 Consequences of Inflexible Prices

So far, I have shown that risk is a significant source of misallocation. An important reason is that, because prices and inventories are chosen in advance, firms are subject not only to the risk of too little demand, but potentially if prices are too low, to missing out on high levels of demand. To investigate this further I compare the model to one where firms choose inventories, I' , and commit ex-post to the market clearing price to sell all inventories. This is the canonical monopolistic pricing model. One can interpret this as a model with labor chosen one period in advance. If I' denotes inventories, prices are:

$$p' = (\theta')^{\frac{1}{\sigma}}\left(\frac{I'}{Y}\right)^{-\frac{1}{\sigma}}$$

The optimal inventories solve:

$$I' = \left[\frac{\widehat{\mathbb{E}} \left[\theta_{t+1}^{\frac{1}{\sigma}} \right] Y^{\frac{1}{\sigma}} \left(1 - \frac{1}{\sigma}\right)}{W(1 + r + \mu)} \right]^{\sigma}$$

This implies that under optimal policy, expected prices are

$$E(p') = \frac{\sigma}{\sigma - 1} W(1 + r + \mu)$$

Unlike the case with predetermined prices, the expected price will be a constant markup over marginal cost. In Figure 5, I compare policies in the benchmark and with flexible pricing. In the flexible price model, I plot expected prices as they vary with realized demand. Note that the policies are very similar when financial frictions are binding but are significantly different when they are not. When $\mu > 0$, the budget constraint of the firm becomes $c \leq m - WI'$. Therefore, since consumption will be very low in these states inventory levels will be quite close to the available cash on hand.[3]

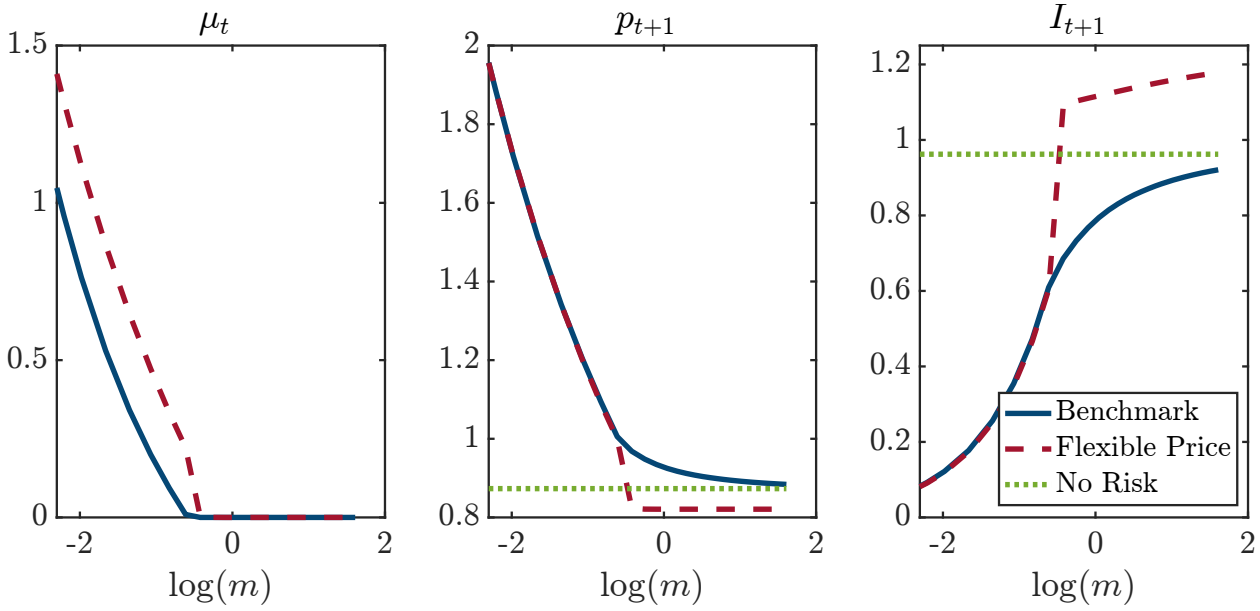
Outside of $\mu > 0$, the policies are significantly different. The expected price becomes identical for all firms under flexible pricing. In addition, there is a significant jump in inventory demand. In fact, with risk aversion and flexible pricing, a firm demands more inventories and plans to sell them at lower prices than the complete insurance counterfactual with one-period-ahead pricing.

4 Empirical Analysis

In this section, I validate firm behavior implied by the model. The model implies that uninsurable risk leads to higher prices, higher markups, and lower inventory investment. Using balance sheet data for U.S. non-financial corporations between 2006 and 2019, I will show that firms facing greater idiosyncratic risk have lower inventory investment. Moreover, variation in exposure to idiosyncratic risk explains a significant share of markup variation in the cross-section. A model without risk-averse firms would not be successful in replicating the data, as markups would not vary and inventory demand would be unrelated to demand volatility.

[3] Under linear preferences with $c \geq 0$, they would exactly equal cash on hand.

Figure 5: Compare Firm Policies to Flexible Counterfactual for Median θ



4.1 Data

Volatility. Following Gilchrist et al. (2014) and Panousi and Papanikolaou (2012), I proxy idiosyncratic uncertainty with the volatility of residual excess returns of the firm's stocks. Data from returns on stocks comes from the Center for Research in Security Prices (CRSP). I observe end-of-week returns from the first week of 2001 until the last week of 2022. I restrict attention to non-financial firms. Let $r^x_{i,t}$ be the return on the stock of firm i at week t. Define $r^x_{m,t}$ as the market excess return and $r^x_{s,t}$ as the sector average excess return.[4] I estimate the following:

$$r^x_{i,t} = \alpha_i + \beta_{m,i} r^x_{m,t} + \beta_{s,i} r^x_{s(i),t} + \varepsilon_{i,t}$$

Define

$$\sigma_{i,t} = \sqrt{\frac{1}{\tau_D} \sum_{\tau \in t} (\hat{\varepsilon}_{i,\tau} - \bar{\varepsilon}_{i,\tau})^2}$$

as the standard deviation of residuals in the excess return in quarter t, where τ_D is the number of trading weeks in quarter t. I use $\sigma_{i,t}$ as a measure of idiosyncratic returns net of predictable components. It reflects variation in equity returns unrelated to the business cycle, sector, or trend. High levels of $\sigma_{i,t}$ reflect high idiosyncratic uncertainty.

[4]Sector is defined at the 3-digit NAICS level.

Balance Sheet. From Compustat Fundamentals Quarterly, I compute measures of investment in physical capital and inventories. As in returns data, I restrict attention to non-financial firms and observations from 2001:Q1 to 2022:Q4. I measure physical capital using the perpetual inventory method. I define inventories as total inventories, that is the measure includes inventories in final goods, materials used in production, and work in progress.

Executive Ownership. As firms in the sample are publicly traded, decision makers such as firm executives are not necessarily as exposed to volatility in returns as they may not own their firm's stock. In the model I present, entrepreneurs are the sole owners of their firms' equity. Following Panousi and Papanikolaou (2012), I use a Compustat supplement on executive ownership to identify firms with managers with an ownership stake in their employer. Public firms must report the ownership stake of their executive officers to the Securities and Exchange Commission. Define $S_{i,t}$ as the share of outstanding stock owned by executives. Due to data quality, only data from 2006 to 2019 are available for executive ownership.

Summary Statistics. Table 4 presents summary statistics for the merged panel of firm volatility with balance sheet and ownership information. I assume that executive ownership is constant within a year. Overall, 50 percent of firm-quarter observations have executives with stock ownership, i.e. $S_{i,t} > 0$. Executives own on average 3 percent of outstanding shares. Note that at the 90th, 95th, and 99th percentiles the values for $S_{i,t}$ are 7, 13, and 35 percent respectively. There is significant dispersion and skewness in executive ownership.

Table 4: Summary Statistics

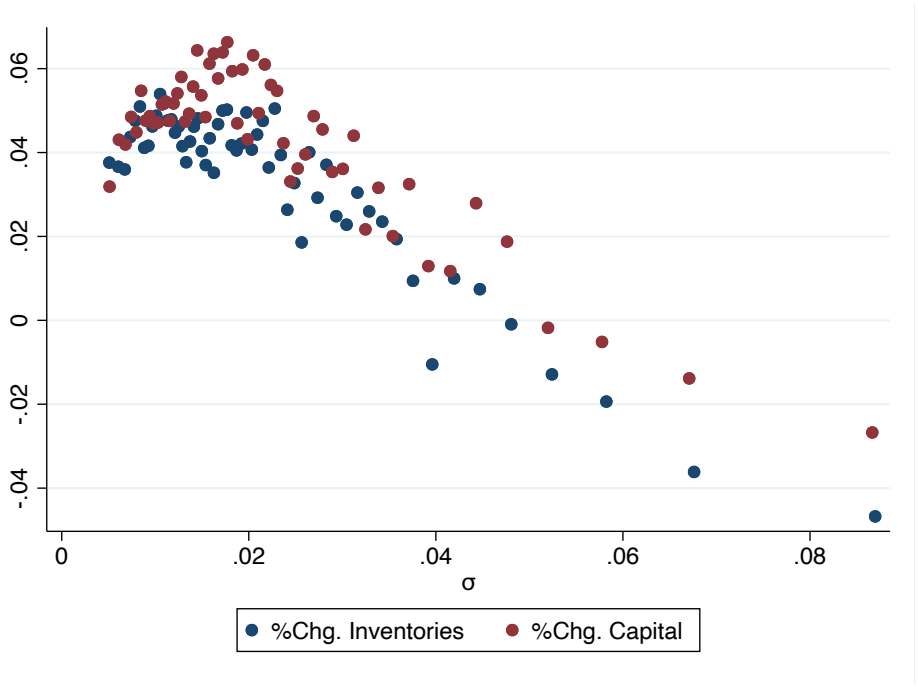
Variable	Mean	SD	25th pct	Median	75th pct
$\frac{\Delta_4 \text{invt}}{\text{assets}}$	0.03	0.22	-0.08	0.03	0.15
$\frac{\Delta_4 k}{\text{assets}}$	0.04	0.23	-0.05	0.01	0.10
$\log(\sigma)$	-3.96	0.62	-4.40	-3.98	-3.53
S	0.03	0.06	0.00	0.01	0.02

4.2 Findings

I report two results: (1) high volatility in excess returns dampens inventory investment and more so for firms with shareholder executives and (2) volatility in excess returns and

shareholder ownership explain 10 to 25 percent of markup dispersion. The first observation points to the relevance of risk aversion for inventory investment. The second observation motivates the focus on markups arising due to risk. The two facts together validate the modeling of pricing and output decisions.

Figure 6: Annual Inventory and Capital Growth against $\sigma_{i,t}$



Inventories and Risk. In Figure 6, I plot binned scatter plots of annual inventory investment, defined as $\log(\text{invt}_{i,t}) - \log(\text{invt}_{i,t-4})$, and annual physical capital investment, defined as $\log(k_{i,t}) - \log(k_{i,t-4})$, on idiosyncratic uncertainty.[5] Notice that as firms face more uncertainty investment falls for both inventories and physical capital. To illustrate the relevance of risk for inventory investment more carefully, I estimate the following specification:

$$\log(\text{invt}_{i,t}) - \log(\text{invt}_{i,t-4}) = \alpha_{s(i)} + \alpha_i + \alpha_t + \beta \log \sigma_{i,t} + \gamma \mathbb{I}\left(S_{i,t} > 0\right) \log \sigma_{i,t} + \Gamma' X_{i,t} + \varepsilon_{i,t}$$

I regress an outcome variable on log volatility $\sigma_{i,t}$ and the interaction between log volatility and an indicator variable for executive ownership. I include sector, firm, and time fixed effects. I also include firm-specific controls in vector $X_{i,t}$. In addition to inventory growth, I also regress annual capital investment rate on the same explanatory variables. Table 5 presents the regression results. Note that higher volatility is associated with lower inventory

[5] Physical capital is measured via the perpetual inventory method.

investment. As observed by Gilchrist et al. (2014), capital investment is dampened. More importantly, the interaction between volatility and executive ownership dampens inventory investment. Therefore, firms with a lot of idiosyncratic risk whose managers are also exposed have lower inventory investment rates. This result extends the findings in Panousi and Papanikolaou (2012) to inventories. Overall, risk aversion can explain the cross-sectional variation in inventory accumulation.

Table 5: Panel Regressions

	Capital	Inventories
$\log(\sigma)$	-0.019***	-0.011***
	(0.002)	(0.002)
$\log(\sigma)$ and $S > 0$	-0.005***	-0.006***
	(0.001)	(0.001)
Controls	log(sales),log(inv.)	log(sales),log(inv.)
N	69719	70359
R^2	0.31	0.37

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Markups and Risk. The model suggests that markups are due to risk. I have abstracted from sources due to imperfect competition or variable demand elasticity. Next, I illustrate that nonetheless variation in risk explains a significant share of dispersion in markups. Measuring markups using only revenue and cost data is fraught as noted by Bond et al. (2021). However, we can still use the ratio of sales to cost-of-goods-sold (COGS) from Compustat to recover sectoral markup dispersion. This is done under the assumption that output elasticities within sectors are identical.[6] Dispersion in log sales to COGS is the dispersion of markups within a sector.

To decompose the dispersion, I run the following fixed-effect regression:

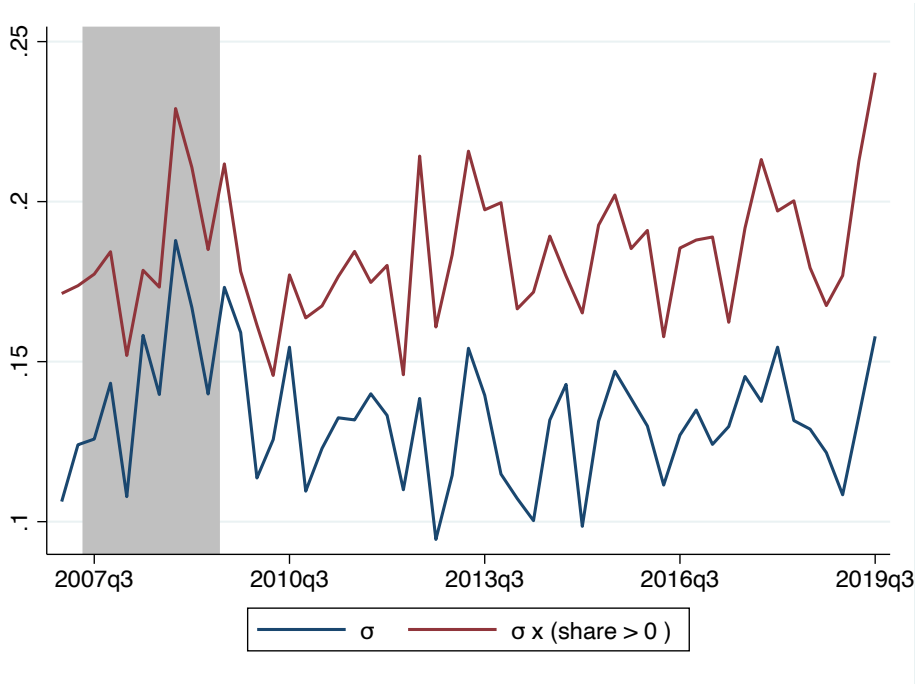
$$\log\left(\frac{\text{sales}_{i,t}}{\text{cogs}_{i,t}}\right) = \alpha_{s(i)} + \alpha_t + \alpha_{\sigma,t}(i) + \varepsilon_{i,t}$$

This regression takes out the sector and time fixed effects, as in the specification above. In addition, $\alpha_{\sigma,t}(i)$ identifies the quintile in $\log \sigma_{i,t}$ to which the firm belongs at time t. Figure 7 plots the standard deviation of $\alpha_{\sigma,t}(i)$ relative to standard deviation of $\log\left(\frac{\text{sales}_{i,t}}{\text{cogs}_{i,t}}\right)$ over

[6] Meier and Reinelt (2024) measure markup dispersion similarly.

time.[7] In addition, I include the same fixed effect for $\log \sigma_{i,t}$ but also interacted with $S_{i,t} > 0$, i.e. identifying exposed executives. Note that volatility of returns, and especially when they are interacted with exposure of executives, explains between 10 and 25 percent of markup dispersion.

Figure 7: Markup Dispersion Explained by Risk



5 Risk and Supply Chain Crisis

In this section, I use the model to study the implications of supply chain crises. During the COVID-19 pandemic and its aftermath, disruptions to the supply chain impacted prices, markups, and stock-out rates.[8] According to the United States International Trade Commission, at the peak of the pandemic 20 percent of shipments by major container shipping firms were canceled.[9] The Census Bureau reports in the Small Business Pulse Survey, which surveys firms weekly between 2020 and 2022, that at the outset of the pandemic 40 percent of firms experienced delays in deliveries from foreign and domestic suppliers.[10] This level

[7] Due to restrictions on markup estimation from balance-sheet data, dispersion is conditioned within a sector.

[8] On stock-outs and prices see Cavallo and Kryvtsov (2023), on markups see Alvarez et al. (2025).

[9] Report accessible at: https://www.usitc.gov/research_and_analysis/tradeshifts/2020/special_topic.html.

[10] Survey accessible at: https://portal.census.gov/pulse.

persisted throughout the survey: the fraction of firms reporting delays ranged between 39 and 65 percent between 2020 and 2022.

In the context of my environment, I model a supply chain crisis as an increase in the portion of inventories that are canceled. Formally, let $\lambda_t(\theta, m)$ be the share of inventories that fail to arrive at time t for a firm with demand θ and cash on hand m, where

$$\lambda_t(\theta, m) = \begin{cases} \lambda_t & \text{with prob. } \eta \\ 0 & \text{otherwise} \end{cases}$$

At the beginning of period t the firm fails to receive $\lambda_t \in [0, 1]$ of their previously ordered inventories with probability η. Exposure to the supply chain crisis is identically and independently distributed across firms and time. While exposure to the crisis is drawn independently by firms, the severity of the crisis, λ_t, is a common shock.[11] The down payment for failed deliveries is returned to the firm, with the interest accrued between periods.[12]

Two parameters are introduced rather than one to capture the risk generated by a supply chain crisis. If λ_t affects all firms with certainty, this setting is isomorphic to one where inventories are taxed at rate λ_t. This does not capture the crucial feature of a supply chain crisis: it leads to an increase in idiosyncratic uncertainty for firms. This shock is motivated by the literature on shipping delays and international trade.[13] Note that this setting is close to models with uncertainty shocks.[14] Rather than have uncertainty shocks changing productivity or demand uncertainty, λ_t implicitly increases the idiosyncratic uncertainty firms are exposed to through stock-outs and low utilization. As was emphasized in previous sections, uncertainty through these sources is critical for understanding pricing behavior of firms.

The first-order condition for prices remains unchanged as the shocks to inventory delivery are conveniently subsumed into the expectation operator. On the other hand, optimal inventory choice changes:

$$\begin{aligned} \left((1 + r_{t+1})(1 - \tilde{\lambda}_{t+1}) + \mu\right) W_t = \left(1 - \tilde{\lambda}_{t+1}\right) \hat{\mathbb{E}}_t \left[\mathcal{X}_{t+1} p' + (1 - \mathcal{X}_{t+1})(1 - \delta_I) W_{t+1}\right] \\ - \hat{\mathbb{C}}ov_t \left(\lambda_{t+1}(\theta', m'), \mathcal{R}_{t+1}\right) \end{aligned} \tag{6}$$

[11] To be clear, in the steady state $\lambda_t = 0$ and hence η as a parameter is irrelevant. Alternatively, a baseline severity (λ_0) and exposure (η) could be included in an alternative calibration of the steady state.

[12] Firms are refunded $W_{t-1} I_t \lambda_t(\theta, m)(1 + r_t)$.

[13] For instance, Alessandria et al. (2023) make the probability of delays in shipments a random variable that is subject to fluctuations.

[14] Examples of papers with uncertainty shocks: Di Tella (2017), Bloom et al. (2018), and Di Tella and Hall (2022).

where as before $\mathcal{X}_{t+1}$ is an indicator of stock-out and

$$\mathcal{R}_{t+1} = \mathcal{X}_{t+1}p' + (1 - \mathcal{X}_{t+1})(1 - \delta_I)W_{t+1}$$

This optimality condition is very similar, with two additions. First, $\tilde{\lambda}_{t+1} = \hat{\mathbb{E}}_t(\lambda_{t+1}(\theta', m'))$ is the average delivery failure rate for inventories under the risk-neutral measure. Second, $\mathcal{R}_t$ is the marginal value of inventories, ex-post.

In equation (6), the right-hand side reflects the marginal benefit of inventories. The supply chain crisis reduces marginal benefit of inventories in two ways: (1) the average shortfall in inventories $\tilde{\lambda}_{t+1}$ the firm expects, under their risk-neutral measure. (2) the covariance of delivery failure rates $\lambda_t(\theta, m)$ and the return to inventories. The former reflects the manner in which supply shocks can appear as implicit taxes on inventories. Of course, such a tax is not uniform as $\tilde{\lambda}_{t+1}$ will vary across firms due to differences in expected marginal utility of consumption.

Prices in a supply chain crisis are the following:

$$p' = \underbrace{\frac{\tilde{\sigma}}{\tilde{\sigma} - 1}}_{\text{markup}} \underbrace{W_t \left((1 + r_{t+1})(1 - \tilde{\lambda}_{t+1}) + \mu \right)}_{\text{marginal cost}}$$

The marginal cost contains an adjustment for refunds if and when there is exposure to the crisis. Markups are a Lerner index again of expected elasticity of demand $\tilde{\sigma}$. However, the elasticity of demand is defined differently to emphasize the role of the supply chain crisis. This elasticity can be expressed as

$$\tilde{\sigma} = \hat{\sigma} \frac{1}{1 + \frac{\hat{\mathbb{E}}_t(\mathcal{X}_{t+1}\lambda_{t+1}(\theta', m'))}{1 - \hat{\mathbb{E}}_t(\mathcal{X}_{t+1})}} \tag{7}$$

If $\lambda_t = 0$, then $\tilde{\sigma} = \hat{\sigma}$, which is expressed as

$$\hat{\sigma} = \sigma \frac{\hat{\mathbb{E}}_t \left(y' | y' \leq (1 - \lambda_{t+1}(\theta', m'))I' \right)}{\hat{\mathbb{E}}_t \left(\min \left\{ y', (1 - \lambda_{t+1}(\theta', m'))I' \right\} \right)}$$

This is the same expected elasticity of demand as expressed in equation (2). Relative to equation (2), I include terms for the possible shortfall in inventories. In equation (7), the supply chain crisis further reduces expected demand elasticity $\hat{\sigma}$ through the final term. This term is a function of the covariance of stock-outs and shock $\lambda_{t+1}(\theta', m')$. The more positive the correlation, the lower the expected elasticity of demand, from the perspective of the firm, and the greater the increase in markups, all else equal.

As the crisis increases risk for firms, the term $\hat{\sigma}$ will be lower relative to the steady state. This channel reflects the general manner in which increased idiosyncratic risk leads to higher markups. One can think of variation in markups due to $\hat{\sigma}$ as reflecting changes due to demand uncertainty. This increases in crises. The additional term in $\tilde{\sigma}$ reflects further amplification due to the non-linearity of the shock and its interaction with stock-outs. This reflects the increase in markups purely due to risk generated by the supply chain crisis and not its interaction with demand uncertainty.

Comparison to Flexible Pricing. Inflexible pricing is a significant source of risk for firms in a supply chain crisis. To see this, in the flexible price counterfactual introduced in Section 3.4, with the same type of crisis, optimal inventories can be expressed as

$$I' = \left[\frac{\widehat{\mathbb{E}}_t \left[\theta'^{\frac{1}{\sigma}} (1 - \lambda_{t+1}(\theta', m')) \right] Y_{t+1}^{\frac{1}{\sigma}} \left(1 - \frac{1}{\sigma}\right)}{W_t((1+r)(1-\tilde{\lambda}_{t+1}) + \mu)} \right]^{\sigma}$$

and prices remain on average with a markup of $\frac{\sigma}{\sigma-1}$, identical to the flexible markup under CES demand. Overall, the supply chain crisis maps to an increase in demand uncertainty. From this one can conjecture that a demand shock would produce similar results, with the caveat that it is a demand shock with heterogeneous effects. Hence, to capture the pure supply side effect of a supply chain crisis one ought to move beyond flexible pricing and give serious consideration to the consequences of price setting.

Timing. I assume the sequence of λ_t is known to firms. At $t = 0$, $\lambda_0 = 0$ and $\lambda_t > 0$ for $t > 0$ and $\lim_{t\to\infty} \lambda_t = 0$. Shocks are not unexpected on impact as this exercise isolates exactly the increase in idiosyncratic risk as opposed to balance sheet effects. Balance sheet effects would be caused by the failure of inventories to arrive at $t = 0$, which affects firms only by reducing their net worth and mechanically increasing the stock-out rate. Since I focus on risk and on increases in stock-out rates, I consider an anticipated crisis. Lastly, I study a partial equilibrium response: I fix interest rates, and only the goods and labor markets clear. I want to emphasize that the results below reflect one particular industry or a small open economy.

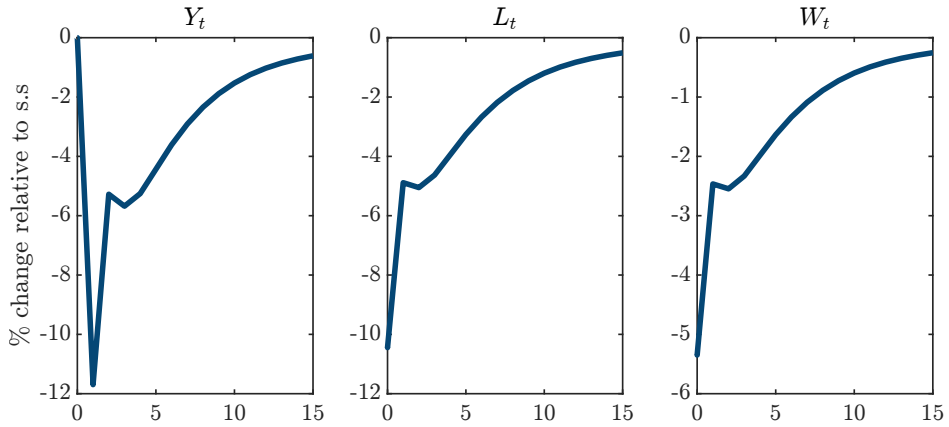
Calibration. I choose $\eta = 0.40$ based on the share of firms in the Census Bureau's Small Business Pulse Survey reporting delays to shipments from either domestic or foreign suppliers. I choose $\lambda_1 = 0.5$ such that on average 20 percent of shipments do not arrive, to match the peak figure in the USITC report. Finally, I assume $\lambda_{t+1} = \rho_\lambda \lambda_t$ and choose $\rho_\lambda = 0.63$

such that the half-life of the crisis is 1.5 years.

5.1 Benchmark Results

Figure 8 plots the path of output (Y_t), inventories (L_t), and wages (W_t), in percentage points relative to the steady state. The crisis implies significant losses for output. In addition, workers producing inventories suffer as a consequence of the fall in wages. The fall in inventories is also significant, implying that stock-out rates are likely to increase throughout the crisis. This is confirmed in the next panel of figures.

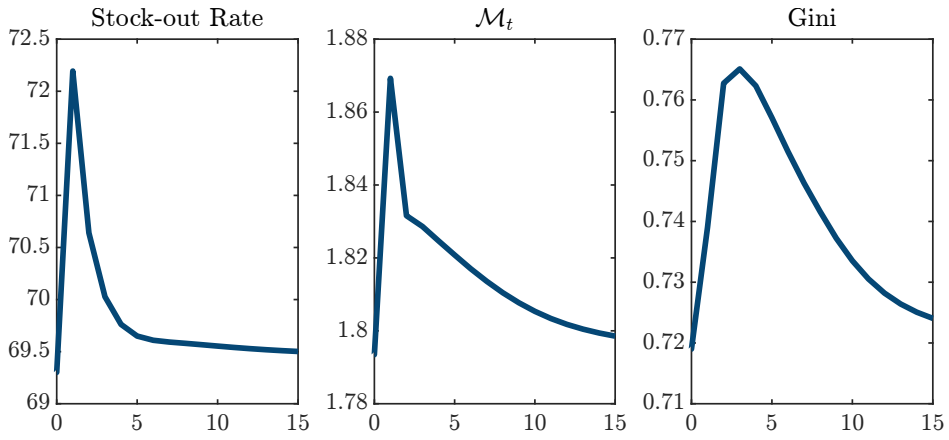
Figure 8: Response of Output, Inventories, and Wages



In Figure 9, I plot the paths of stock-out rates, the aggregate markup $\mathcal{M}_t$, and the Gini coefficient of wealth among firms. As noted, the crisis leads to an increase in stock-outs, by 5 percent at the peak, as was initially calibrated with λ_t. This stock-out increase coincides with an increase in markups; the aggregate markup increases from 1.79 to 1.87. Inequality also worsens: the Gini coefficient increases from 0.72 to 0.76. As the wealth distribution through risk compensation via markups determines the aggregate markup, one can conjecture that the magnitude of the effects on markup and output are likely amplified by the wealth heterogeneity. In fact, stock-out rates return quickly to the steady state values relative to the aggregate markup, which remains persistently above the steady state level, as does the Gini coefficient.

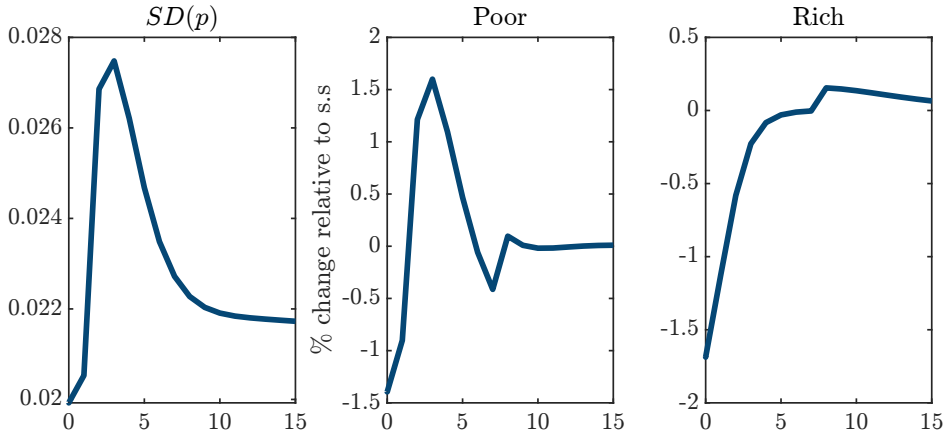
In Figure 10, to study the behavior in the cross-section, I plot the standard deviation of prices and the average price policy by poor and rich firms. I define poor firms as those that are at the borrowing constraint, that is $a' = 0$. Rich firms are otherwise financially unconstrained, $a' > 0$. At the peak effect of the supply chain crisis prices become extremely dispersed. I find that poor firms adjust prices upward during the crisis on average and wealthy firms have relatively flat responses. Much of the price dispersion is being driven by

Figure 9: Response of Stock-out Rates, Markups, and Gini



the differential wealth of firms. This illustrates the manner in which pricing can be used as a precautionary measure. Poor firms care a great deal about stock-outs as their inventory demand is already significantly restricted. Therefore, they want to insure the high degree of profits in bad states of the world. Wealthy firms behave closer to risk-neutral firms; their pricing policy reflects greater responsiveness to the aggregate state. Expecting a fall in output, they lower their price. All firms on impact have lower prices to reflect aggregate output, Y_t, falling and shifting demand everywhere lower.

Figure 10: Moments of the Price Distribution



5.2 Counterfactuals

In this section, I consider two counterfactual exercises to understand the role of risk and pre-determined price setting for outcomes in a supply chain crisis. To understand the role of risk, I will repeat the exercise for an economy with risk-neutral firms without any financial frictions. To understand the consequences of price setting in the benchmark economy, I

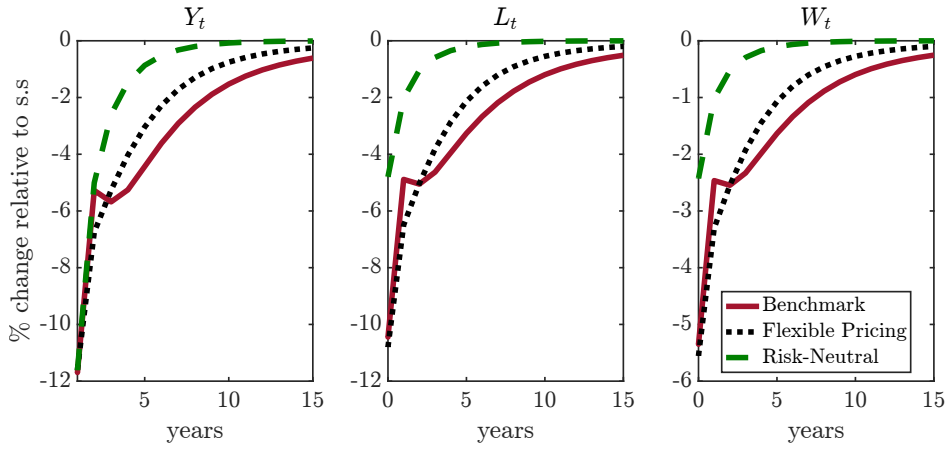
repeat the crisis in an economy with flexible pricing, keeping preferences and financial frictions untouched. For each counterfactual, I recalibrate λ_t such that the peak impact on Y_t is identical to the benchmark response.

Table 6: Counterfactual: Steady States

	Benchmark	Risk-Neutral	Flexible Pricing
Y	0.58	36%	34%
Z	0.98	1.2%	5%
$\mathcal{M}$	1.79	-4.0%	-7%
Stock-out	69%	61%	–
Gini	0.72	–	20%

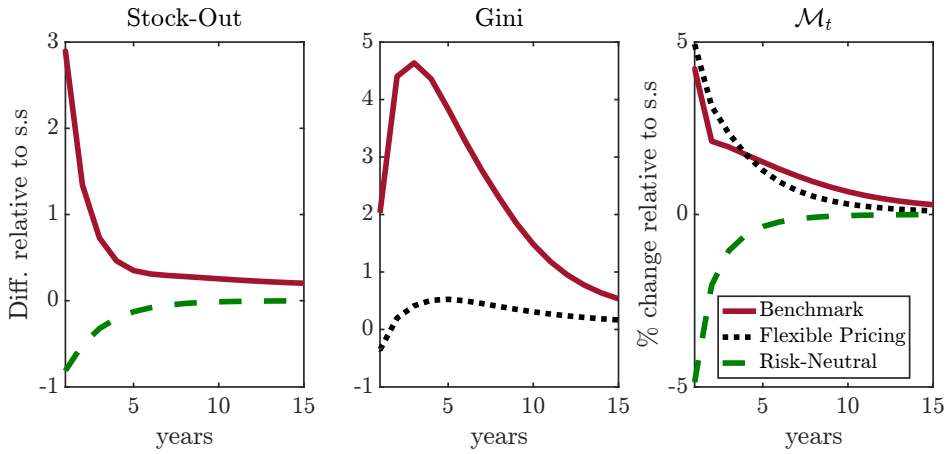
To begin with, I compare the steady state properties of the different environments in Table 6. The first column presents values for the moments of interest in the benchmark economy and the subsequent columns present the percentage difference from the counterfactual economy. Both risk-neutral and flexible-price economies have significantly higher output. This is due primarily to greater aggregate productivity and lower aggregate markups. This is a consequence of the role of risk in the benchmark economy. For the supply chain crisis, Figure 11 compares the path of output, inventories, and wages for the benchmark economy and the two counterfactual exercises. Figure 12 compares the stock-out rates, Gini coefficient, and aggregate markups across models.

Figure 11: Counterfactual: Response of Output, Inventories, and Wages



Compare to Risk Neutrality. From Figure 11, I observe that risk leads to more pronounced and persistent downturns as a consequence of the supply chain crisis. The bench-

Figure 12: Counterfactual: Response of Stock-outs, Gini, and Markups



mark economy takes longer to recover. Moreover, the response in wages and inventories is more than twice as strong. Figure 12 shows that qualitatively without risk, stock-out rates and markups fall. Hence, a major contribution to the differential response in wages is the fall in inventories.

Compare to Flexible Pricing. From Figure 11, recovery with flexible pricing is slower than the risk-neutral benchmark; however, inflexible pricing slows recovery of outcomes significantly. On impact, wages fall similarly in the flexible pricing model and benchmark, suggesting similar movements in markups. This is confirmed in the final panel of Figure 12. While the expected price always carries a markup of $\frac{\sigma}{\sigma-1}$, the standard CES value, the wedge in the firm-level labor share ex-post leads to a time-varying wedge in labor income, which one can label as the aggregate markup.[15]

6 Conclusion

This paper introduces a model of precautionary pricing in which firms facing uninsurable, idiosyncratic demand risk set prices and inventories prior to the realization of demand. Risk aversion induces firms to price as a form of insurance against the risk of stock-outs and surplus inventories. As a result, the model generates dispersion in prices and markups even under constant demand elasticity, and links markup variation to firms' financial positions and exposure to demand volatility.

Quantitatively, the model accounts for key cross-sectional moments in U.S. firm-level data, including the relationship between volatility, inventory investment, and markup dis-

[15]This is akin to the aggregate wedge due to labor in Boar et al. (2022) or Di Tella et al. (2025).

persion. The empirical evidence confirms that firms with greater idiosyncratic risk, especially those with exposed executives, invest less in inventories as predicted by precautionary motives.

The analysis highlights that idiosyncratic risk and incomplete insurance can generate significant misallocation and amplify macroeconomic fluctuations. During supply chain disruptions, increased uncertainty about inventory availability raises markups and prolongs recovery relative to an economy with risk-neutral firms.

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